

(a) Where the noise lives: per-op approximation error

error = true change

corrector

IG (Z $\times$)

midpoint

secant

midpoint

midpoint

freeze

secant

■ GPT-2 small
■ Gemma-2-2B

softmax

1.46

1.12

Q@K

0.66

0.83

MLP act (GELU)

0.72

0.59

GeGLU gate-up

0.68

A@V

0.27

0.15

Norm (LN/RMS)

0.25

0.31

logit softcap

0.01

0.0

0.5

1.0

1.5

EAP op-gap (relative first-order error)

(b) softmax: no closed form — needs Z-step IG

EAP tangent (no integration)

softmax op-gap

10^0

10^{-1}

10^{-2}

10^{-3}

1

2

5

10

20

integration steps Z

● GPT-2 small

● Gemma-2-2B